

Jensen Beach to Jupiter Inlet Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

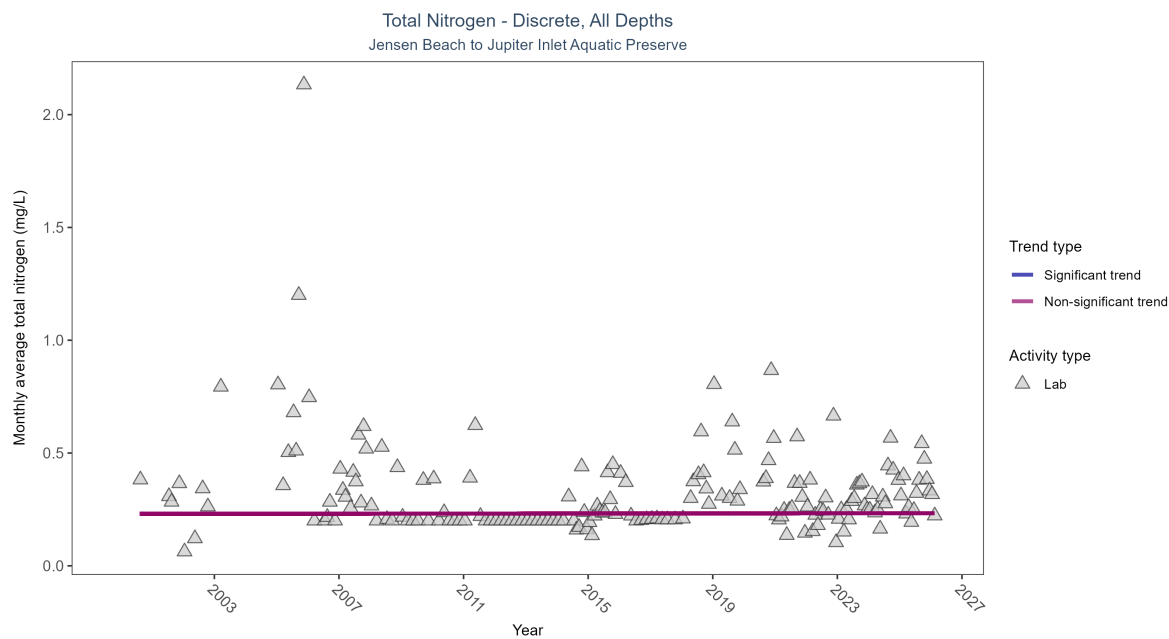


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 2000 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	742	26	2000 - 2026	0.2615	0.0551	0.23023	0.00012	0.3333

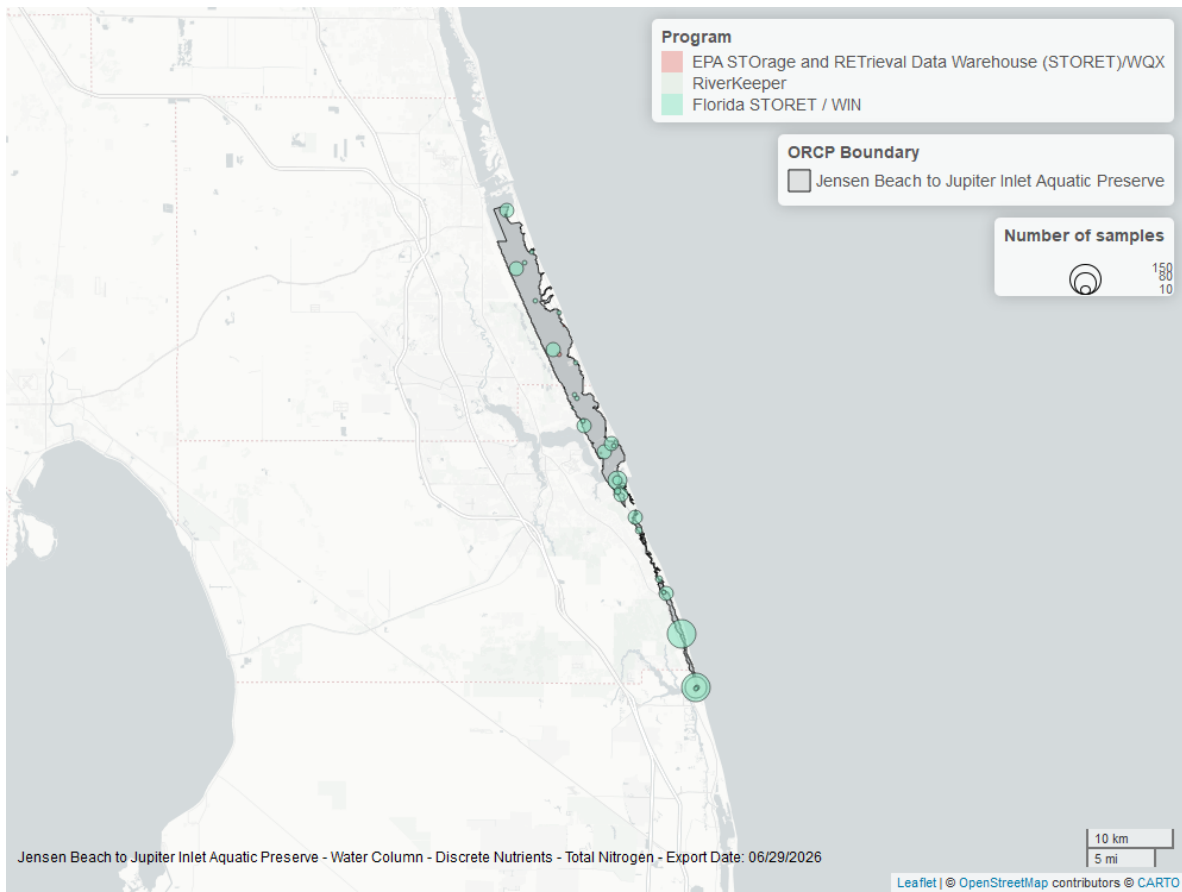


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

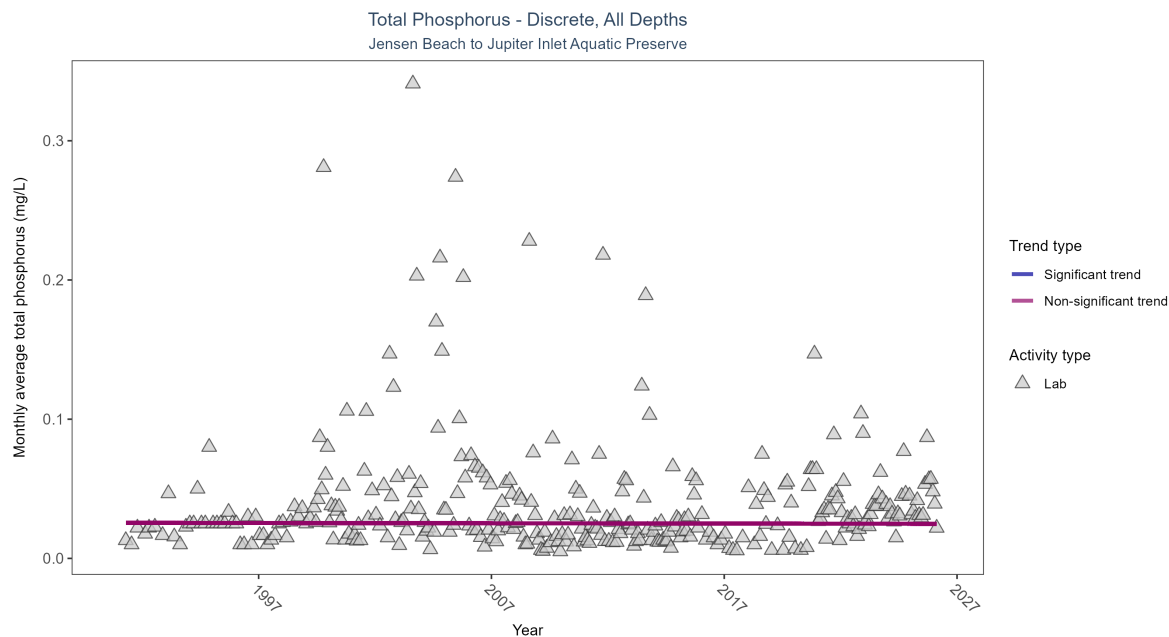


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1991 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1175	36	1991 - 2026	0.027	-0.02114	0.02555	-0.00002	0.8002

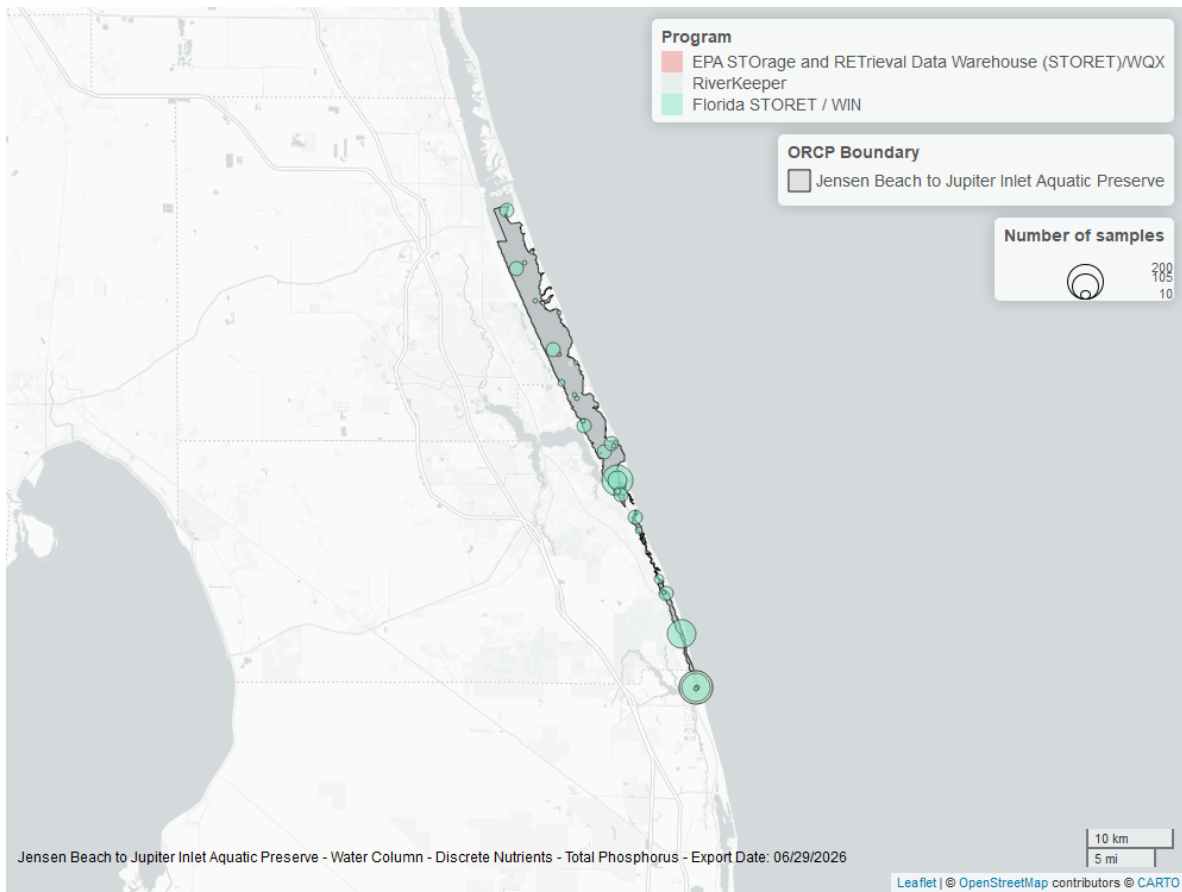


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

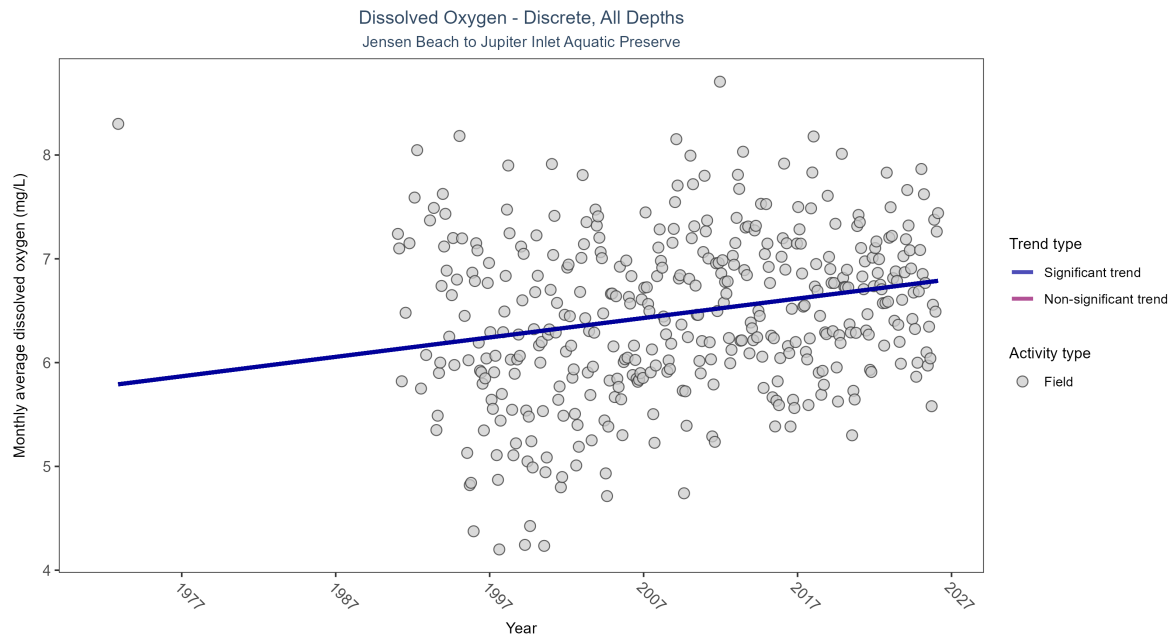


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Monthly average dissolved oxygen increased by 0.02 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	11083	37	1972 - 2026	6.42	0.21627	5.77463	0.01872	0

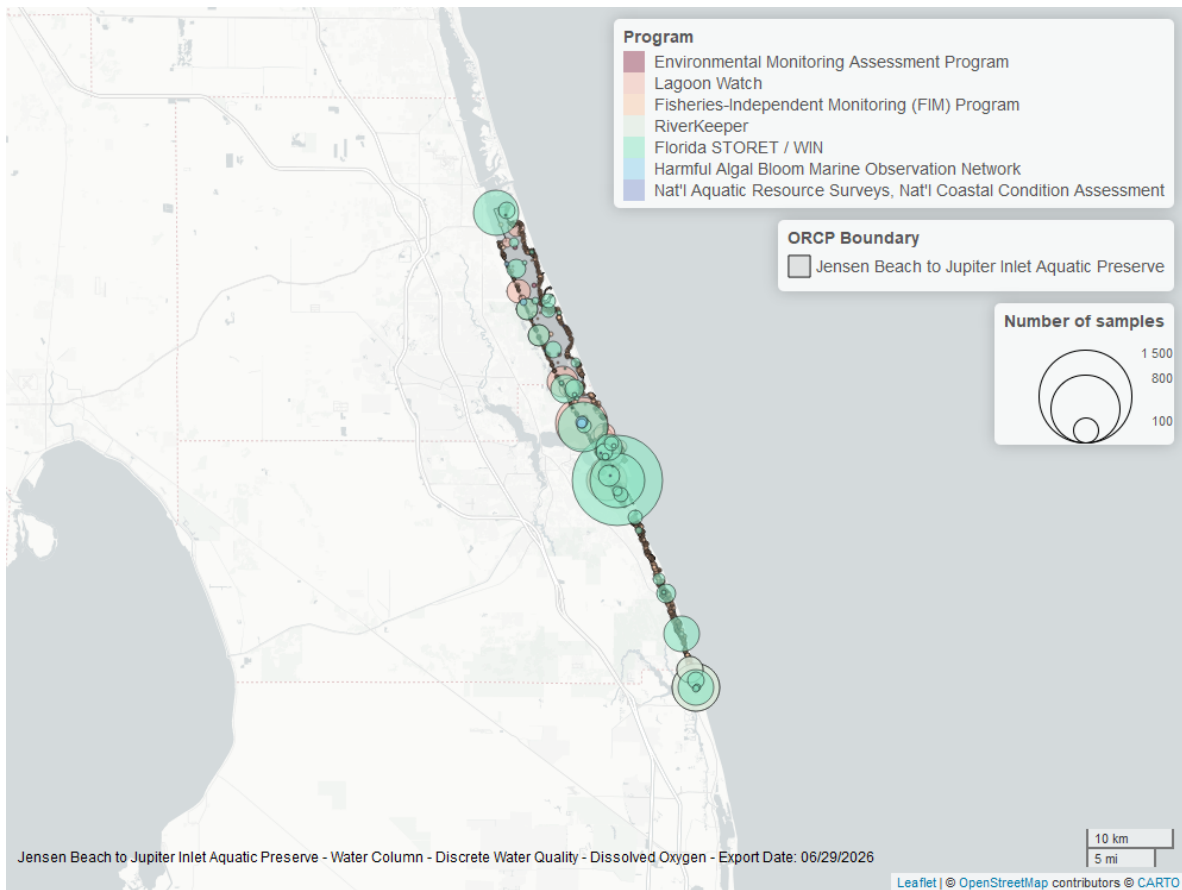


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

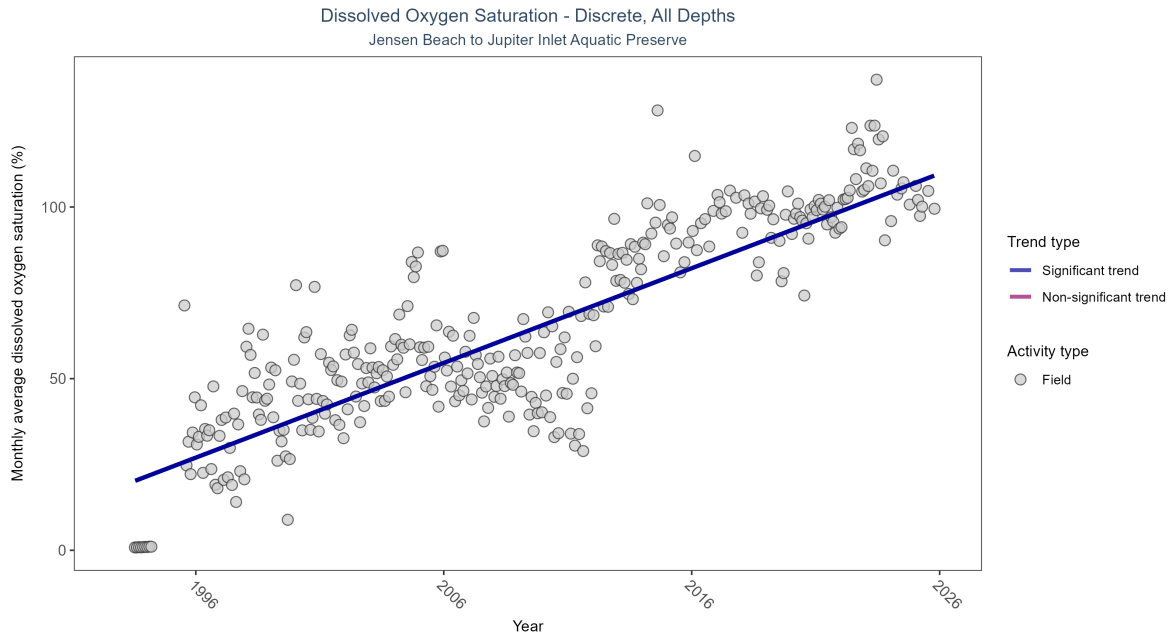


Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 2.76% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	4302	33	1993 - 2025	80	0.68087	18.71729	2.75845	0

\end{table}

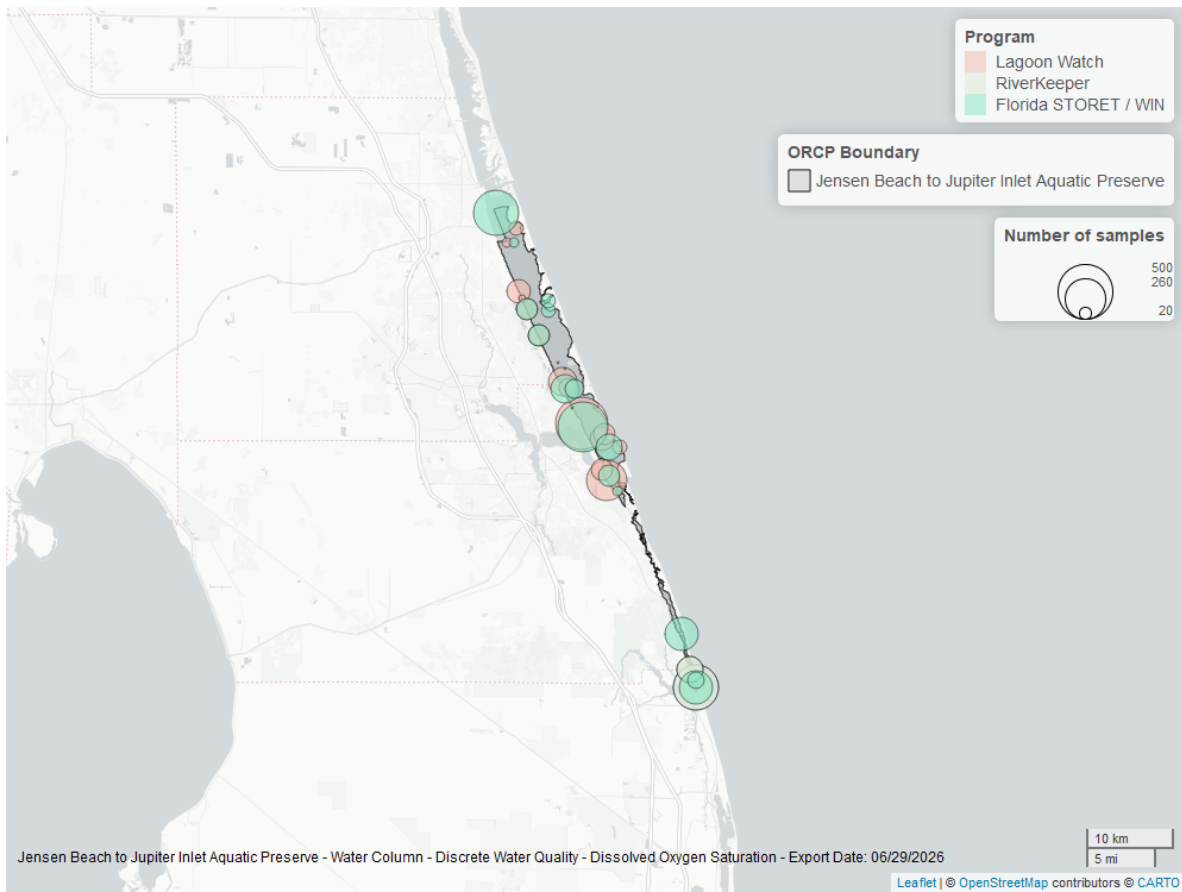


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

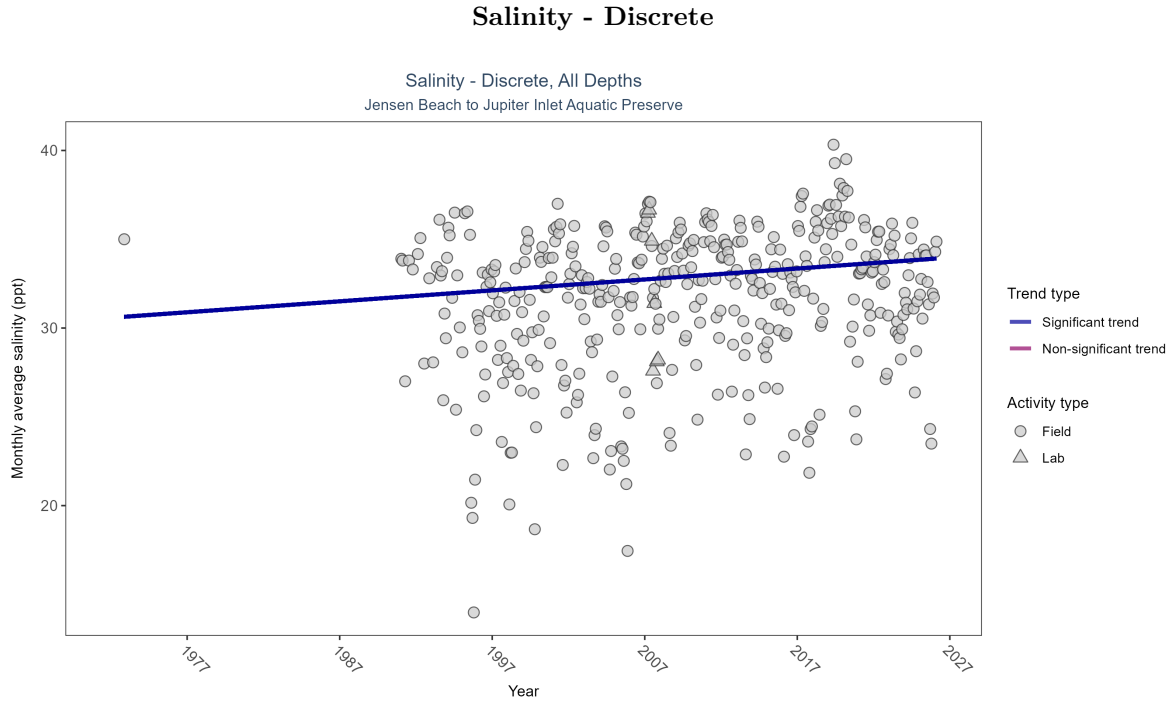


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 4: Monthly average salinity increased by 0.06 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	11371	37	1972 - 2026	32.8	0.1335	30.58017	0.0616	0.0001

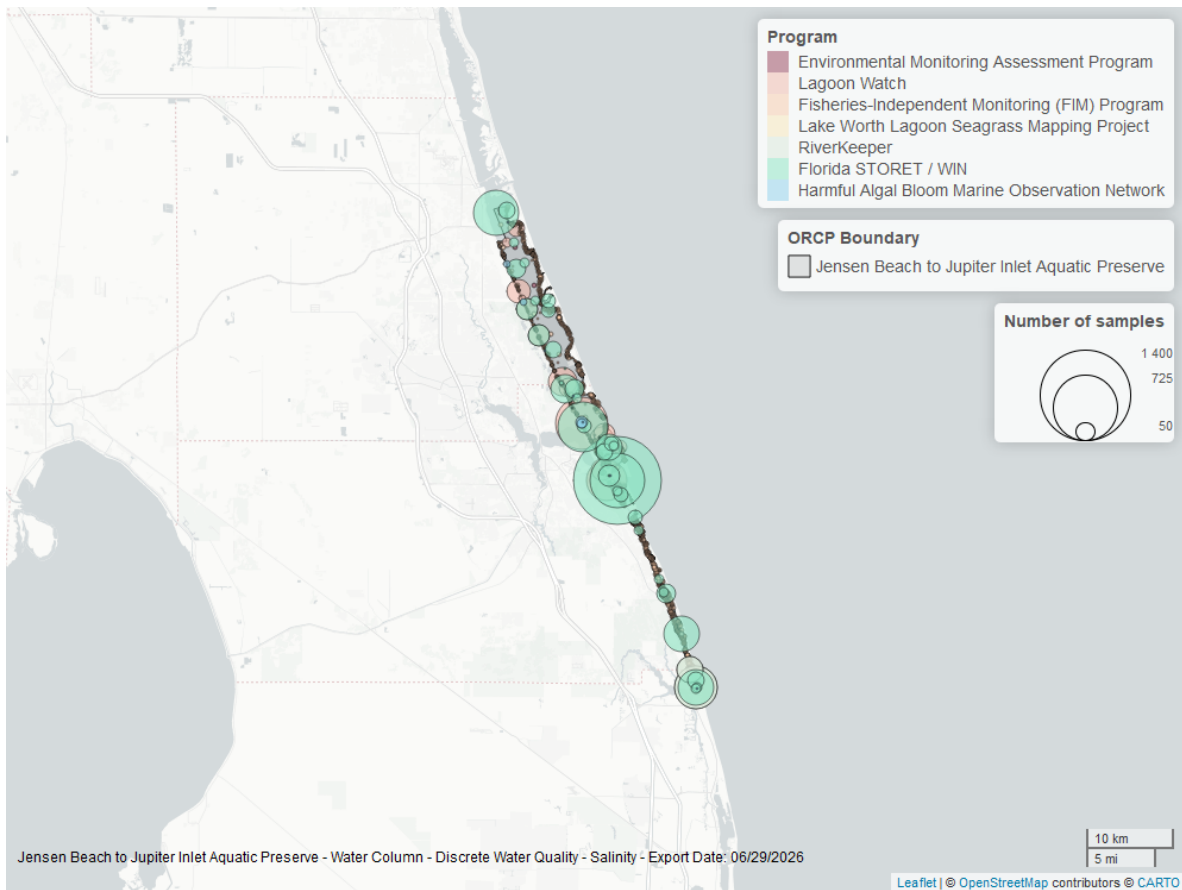


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

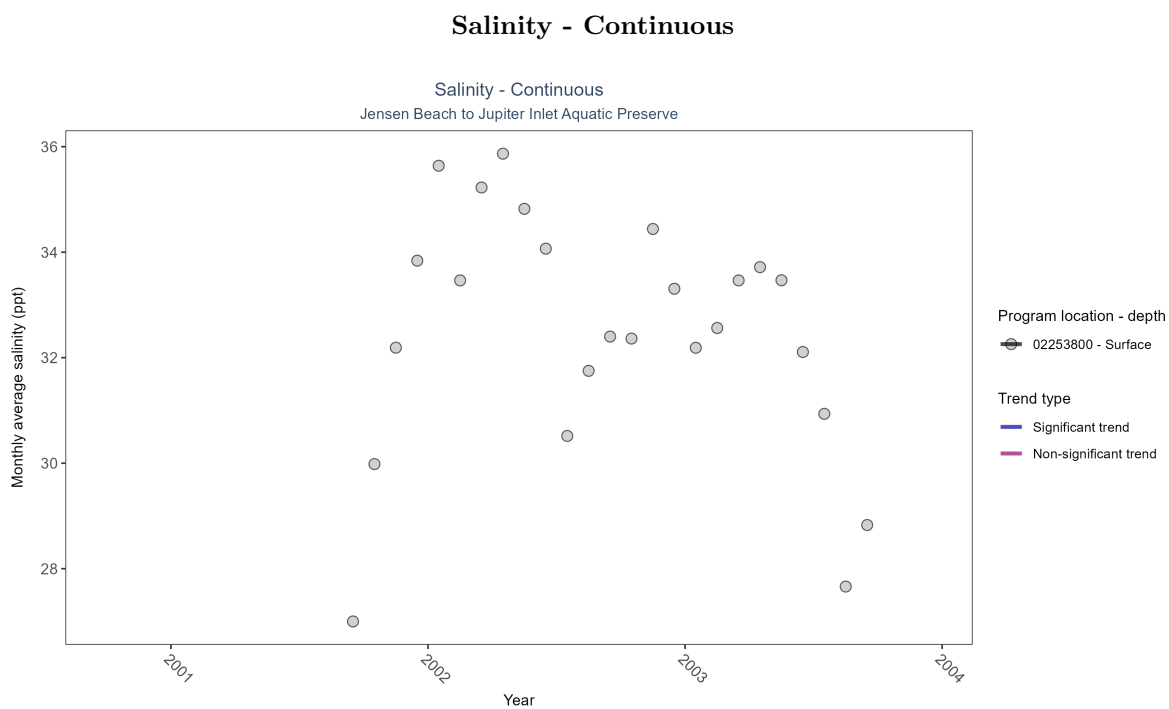


Figure 11: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02253800	Insufficient data	1213	3	2001 - 2003	33	-	-	-	-

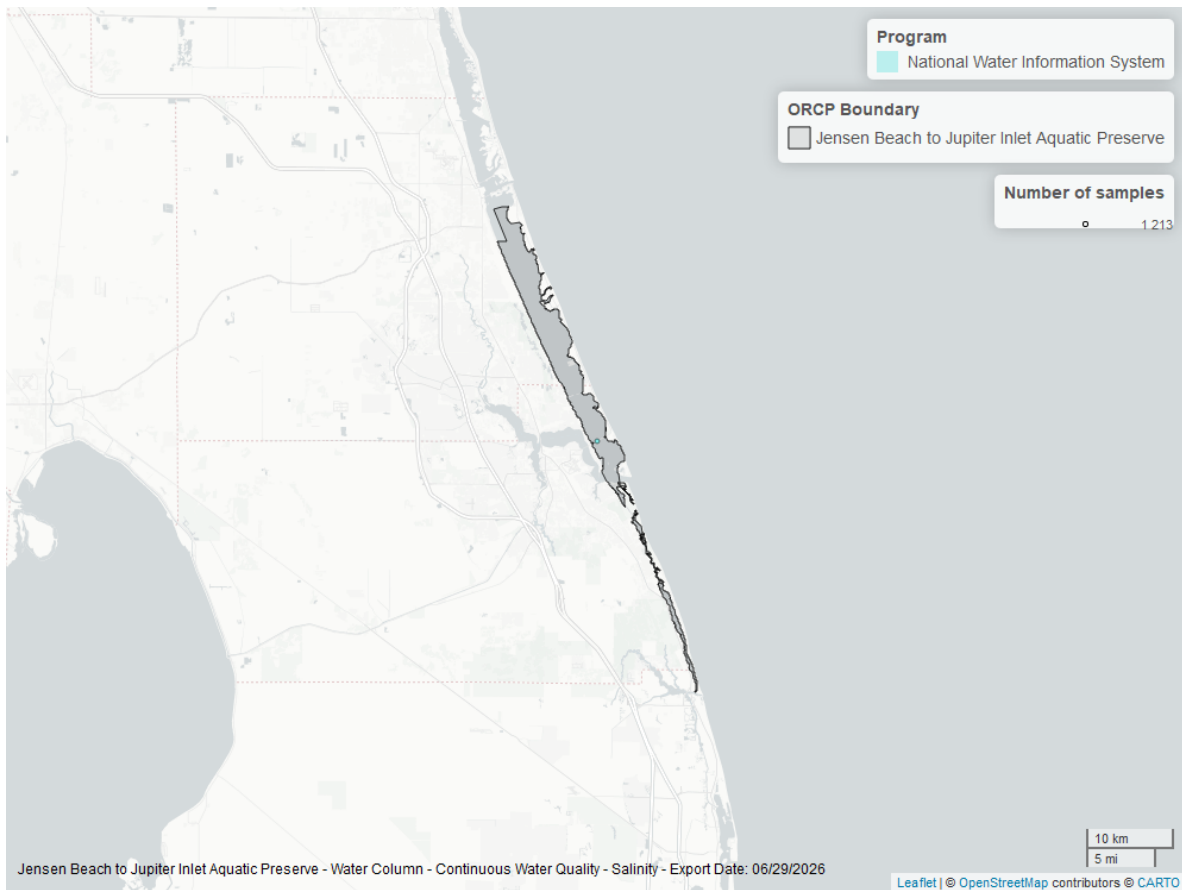


Figure 12: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

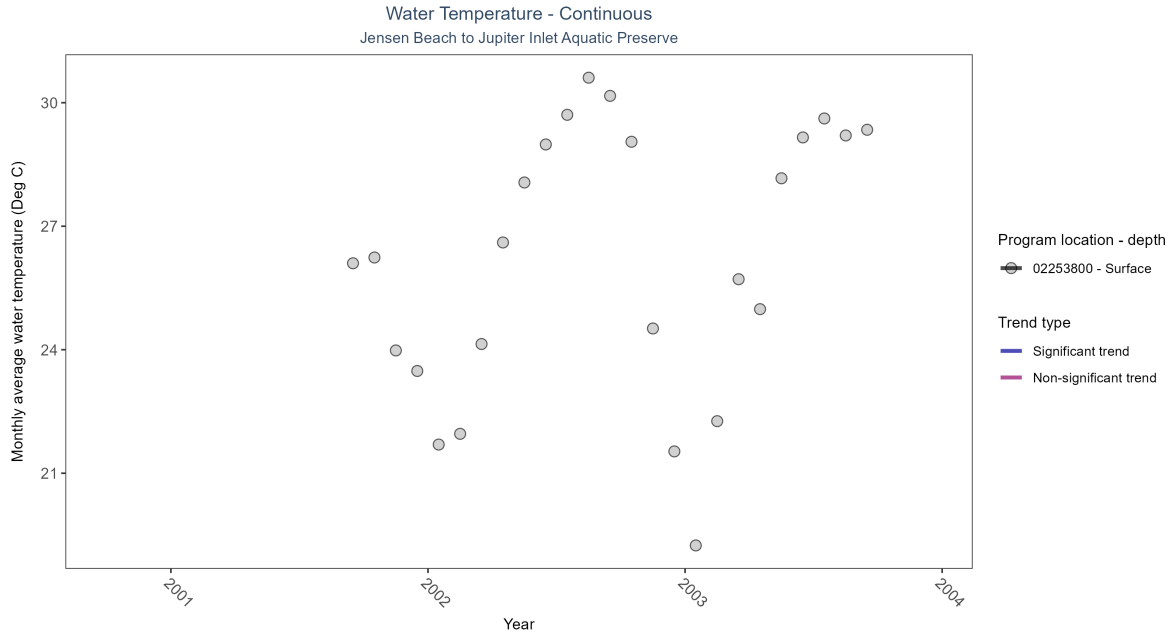


Figure 13: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02253800	Insufficient data	1186	3	2001 - 2003	27.1	-	-	-	-

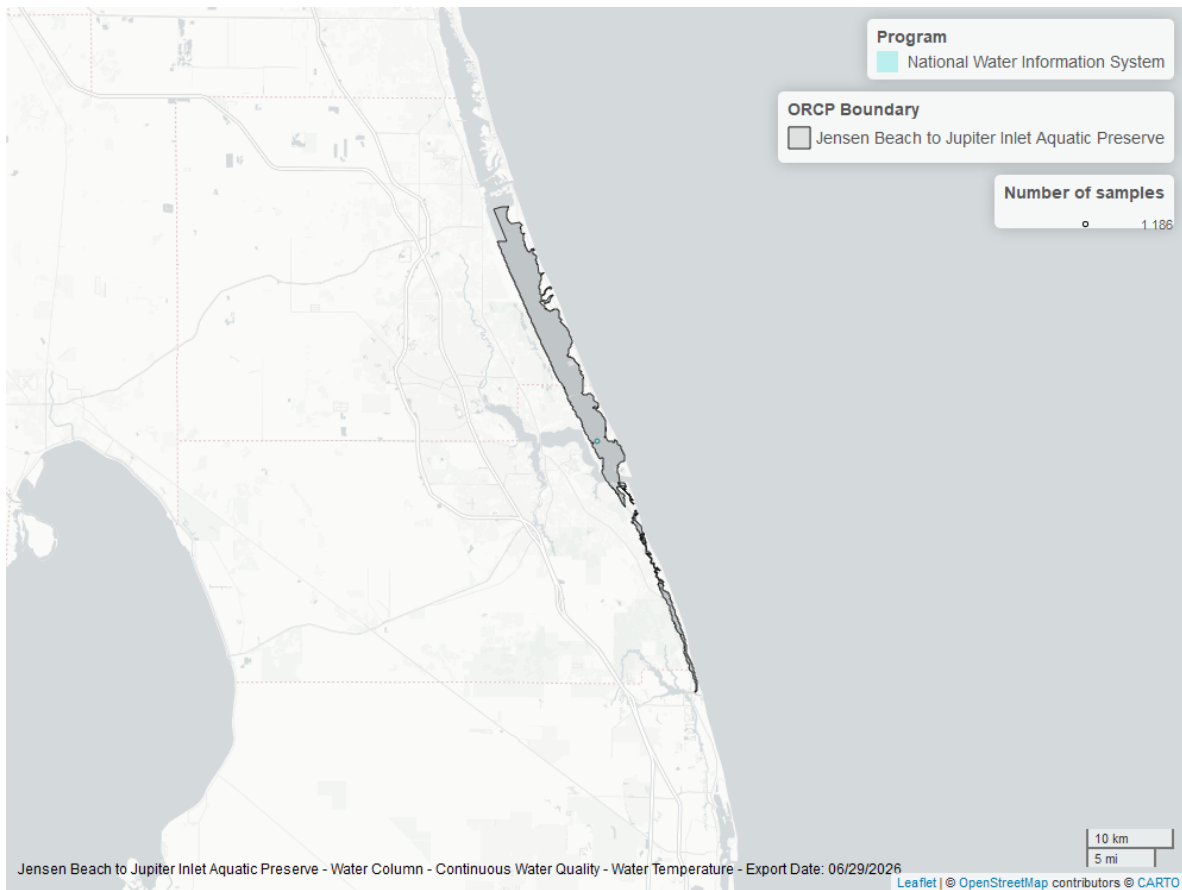


Figure 14: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

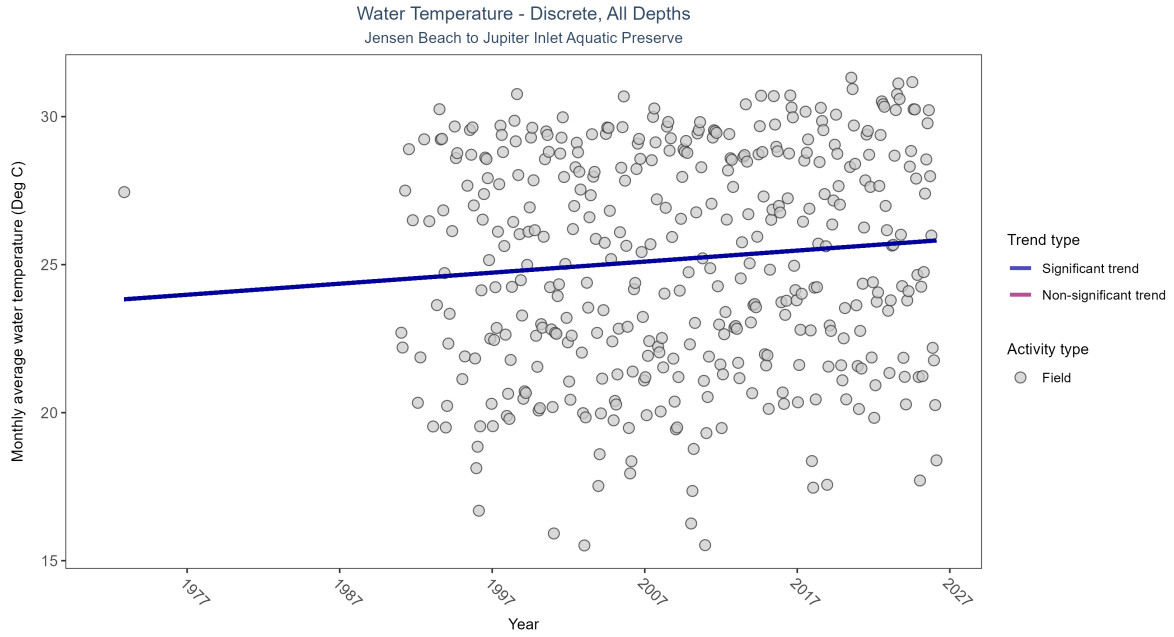


Figure 15: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 7: Monthly average water temperature increased by 0.04Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	11131	37	1972 - 2026	25.5	0.21646	23.79669	0.03731	0

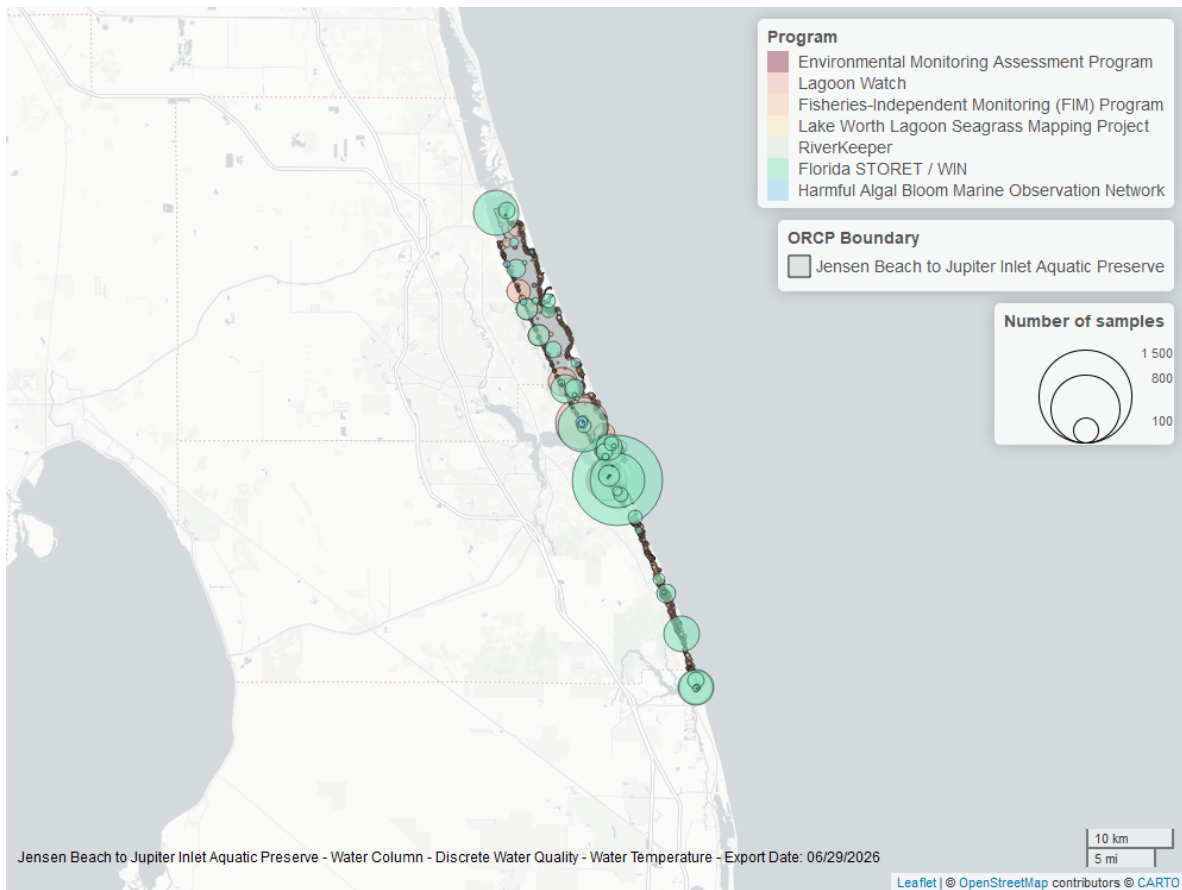


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

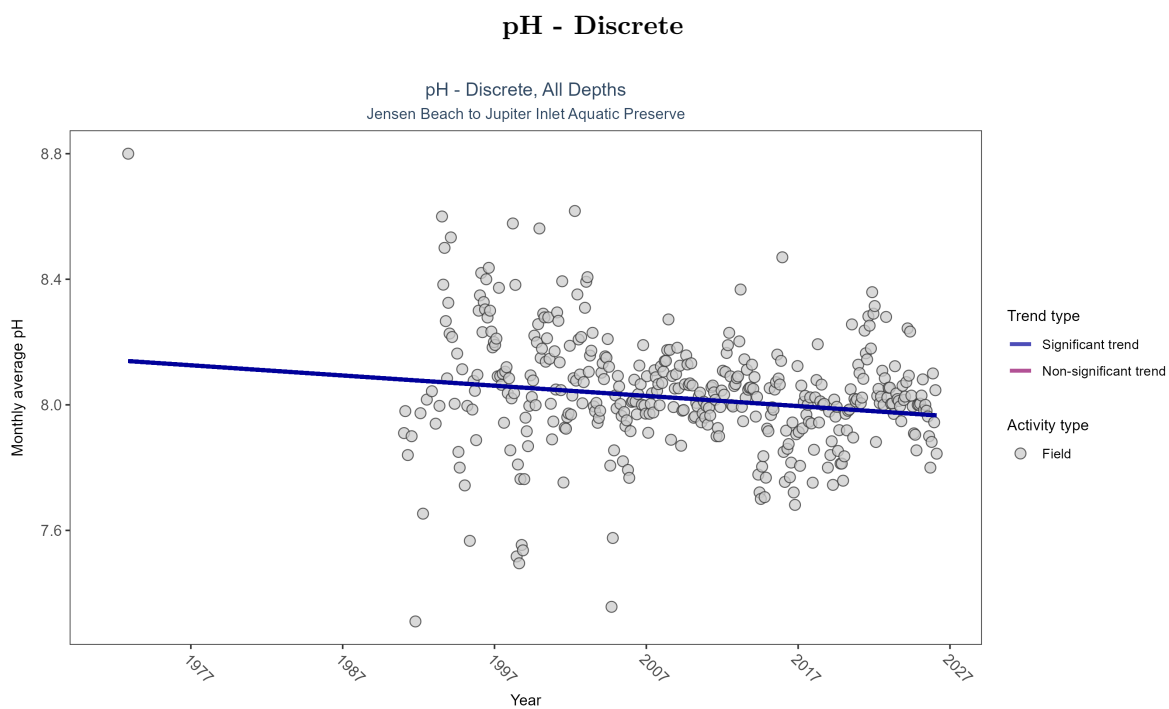


Figure 17: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 8: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	10057	37	1972 - 2026	8.01	-0.15018	8.14225	-0.00325	0

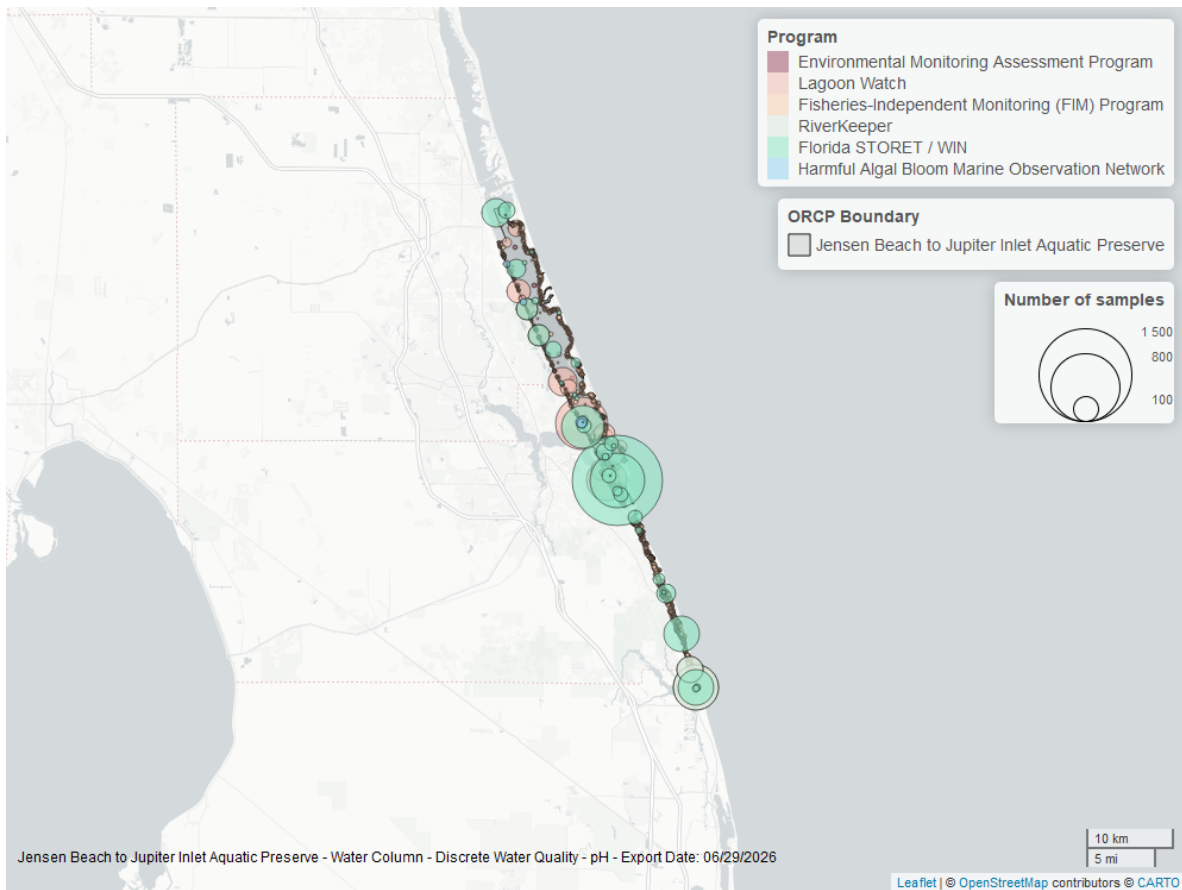


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Discrete

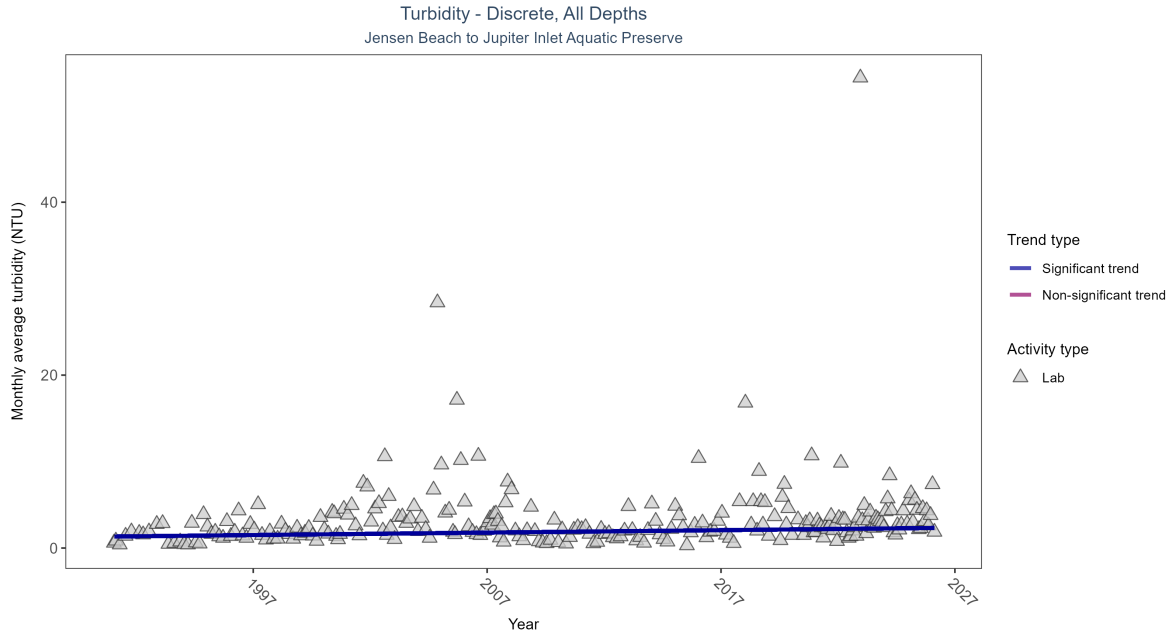


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 9: Monthly average turbidity increased by 0.03 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	986	36	1991 - 2026	2.3	0.13945	1.3314	0.02821	0.0039

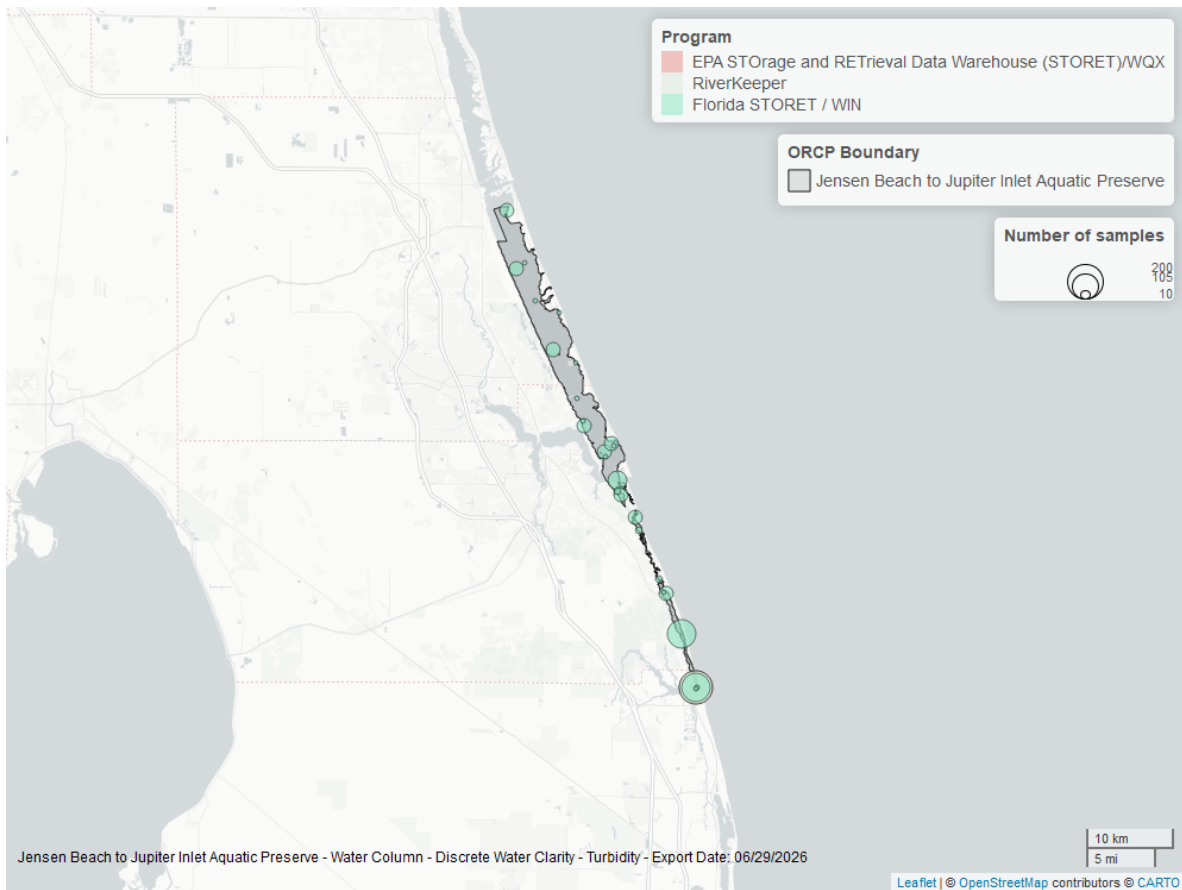


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

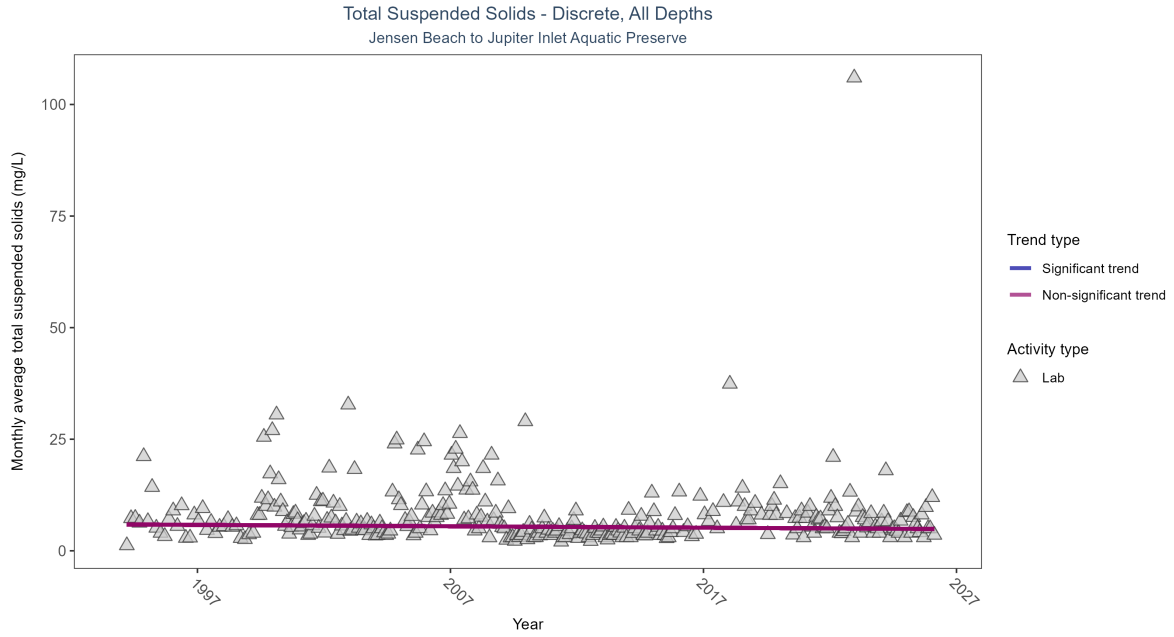


Figure 21: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 10: Total suspended solids showed no detectable trend between 1994 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1303	33	1994 - 2026	6	-0.07224	5.90251	-0.03156	0.0969

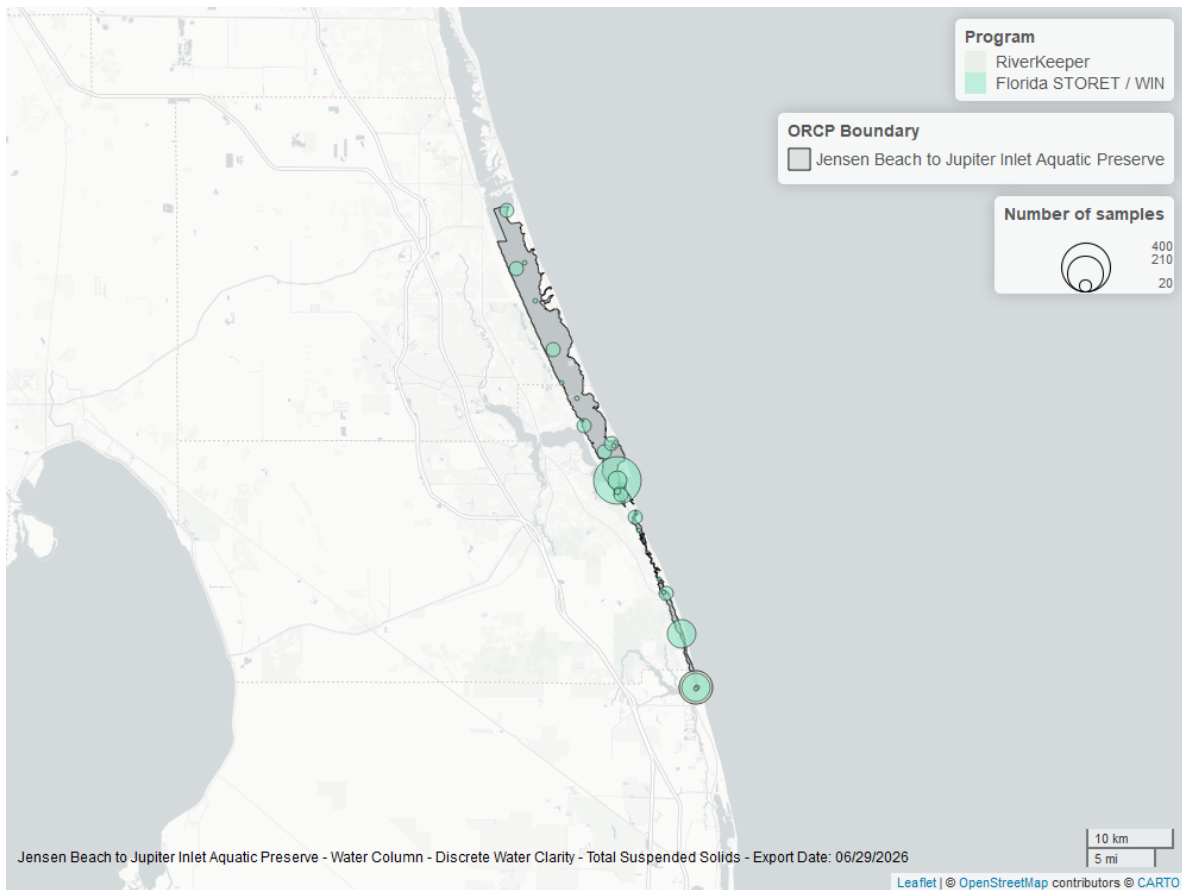


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

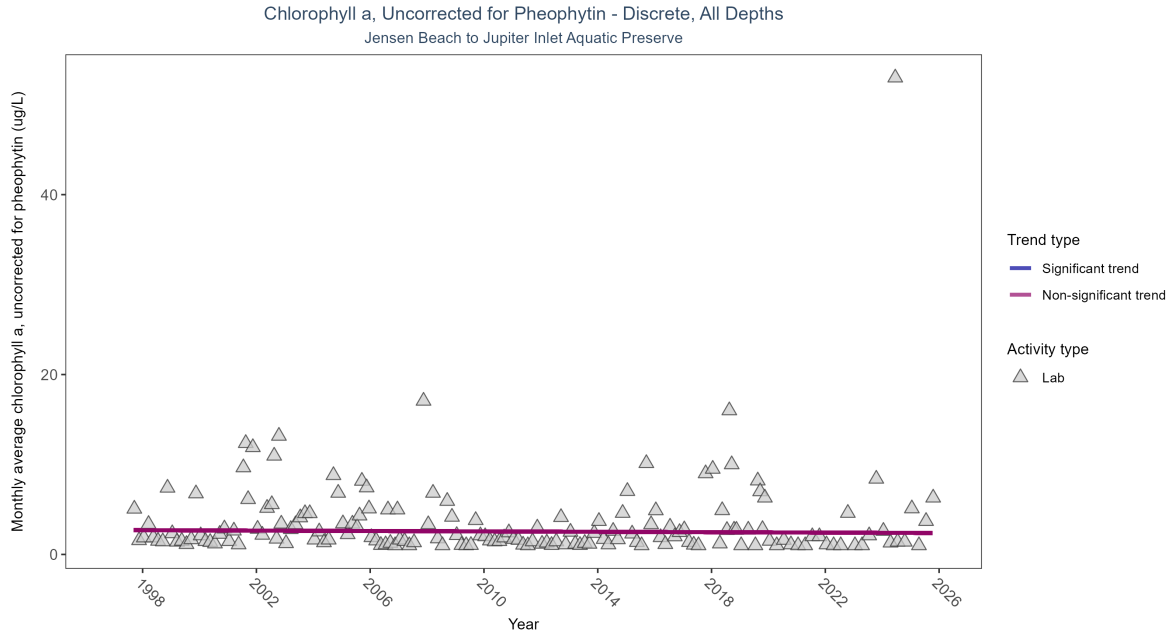


Figure 23: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 11: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1997 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	479	29	1997 - 2025	2	-0.10169	2.70475	-0.01111	0.1899

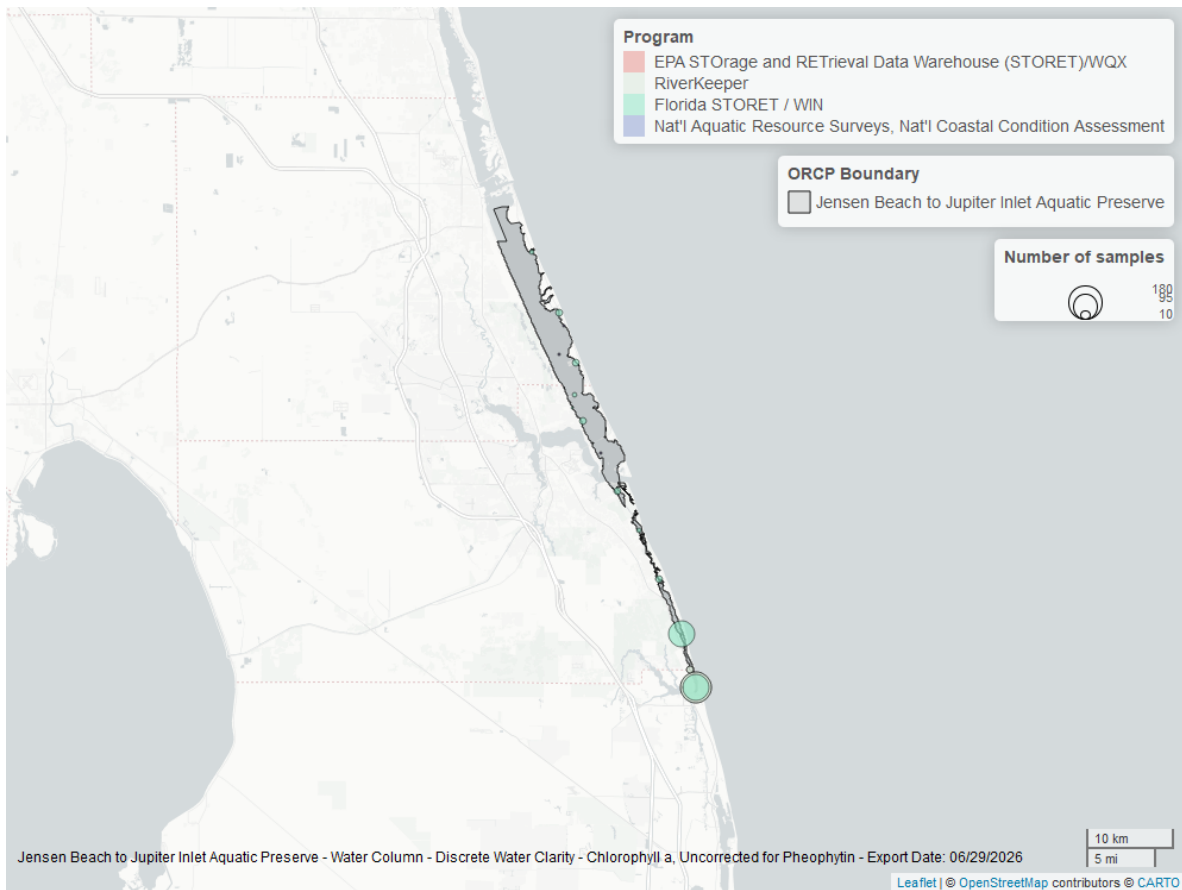


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

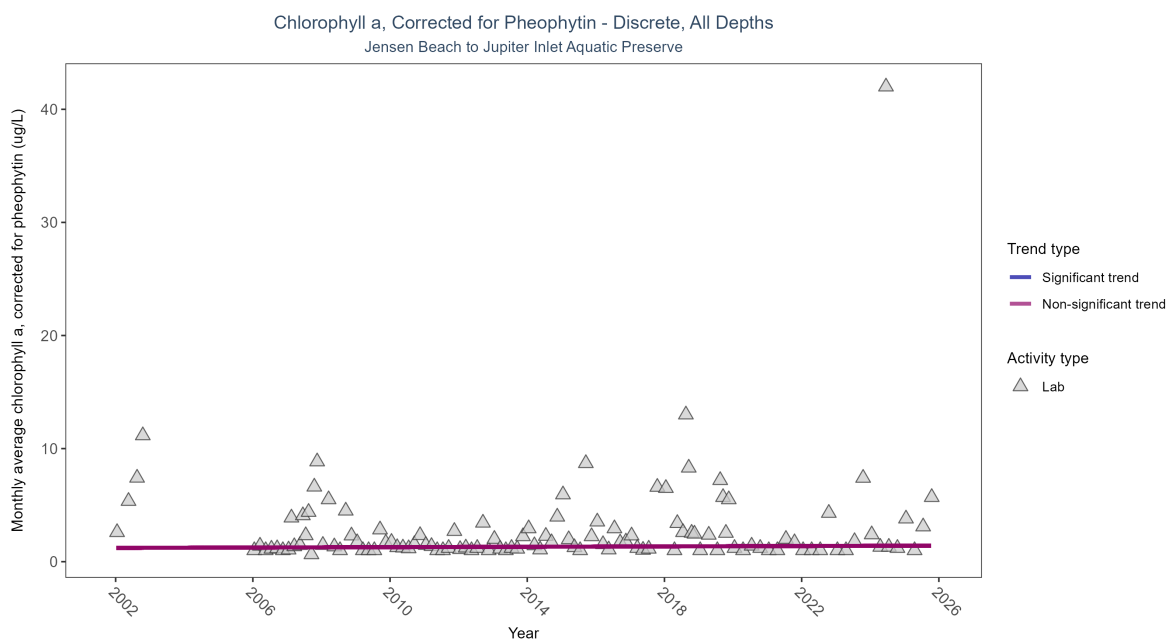


Figure 25: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 12: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	349	21	2002 - 2025	1.4	0.09098	1.21844	0.00833	0.1673

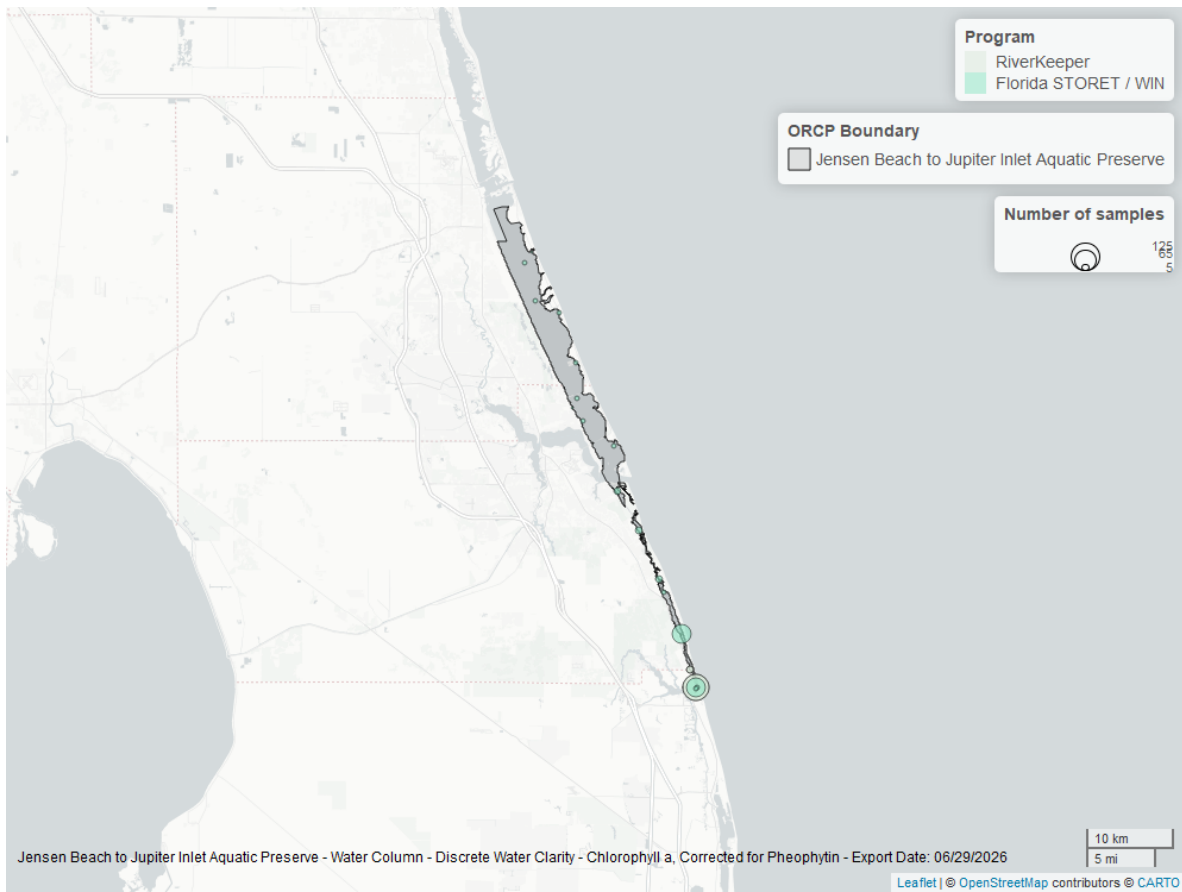


Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

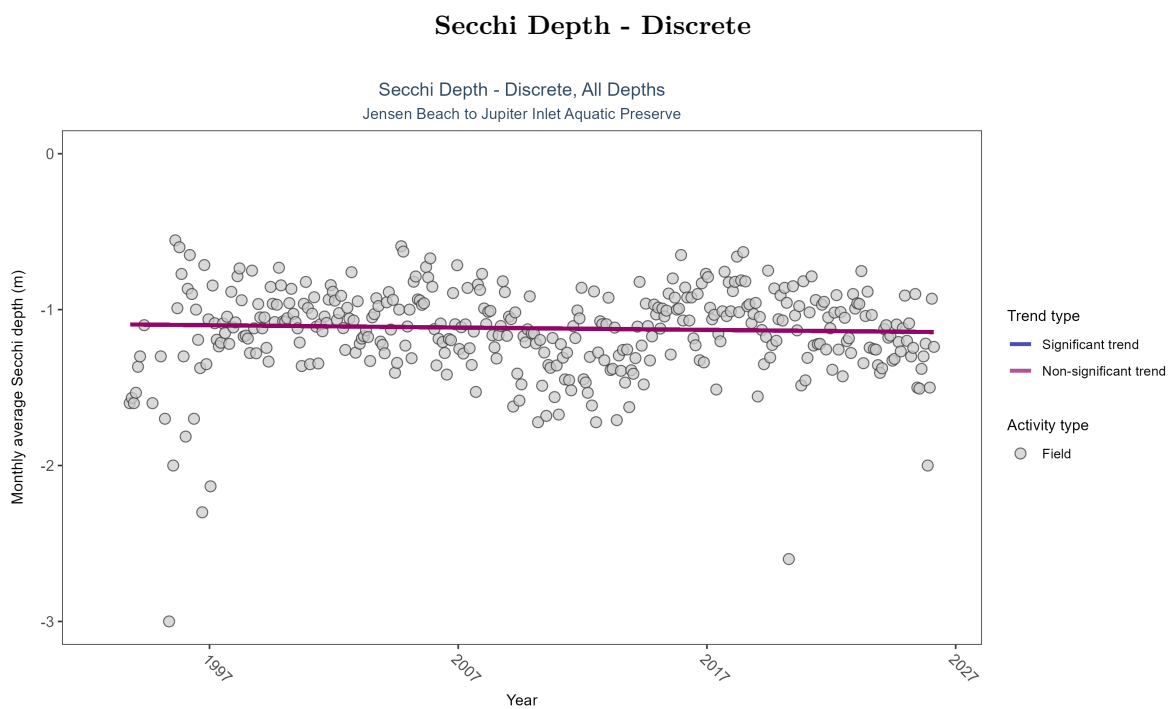


Figure 27: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 13: Secchi depth showed no detectable trend between 1993 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	6588	34	1993 - 2026	-1	-0.03632	-1.09407	-0.0015	0.336

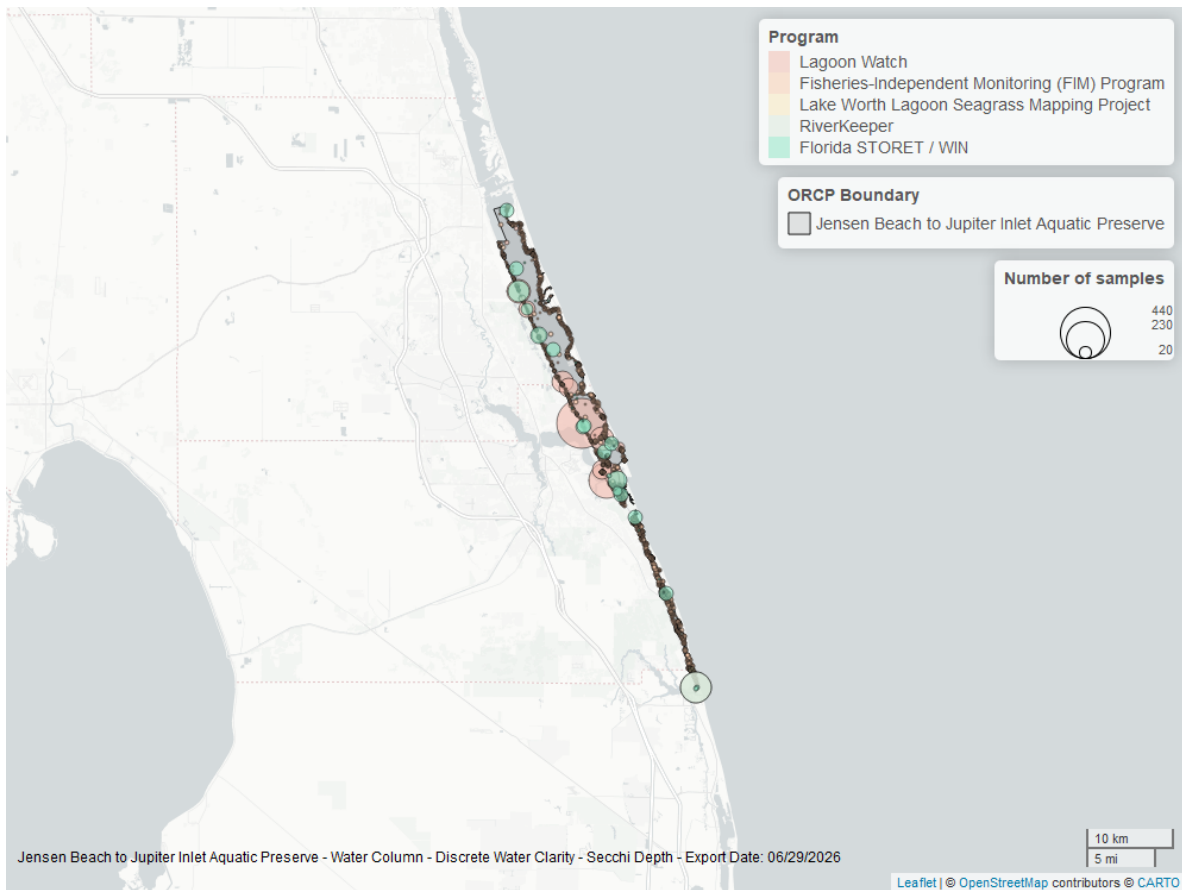


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

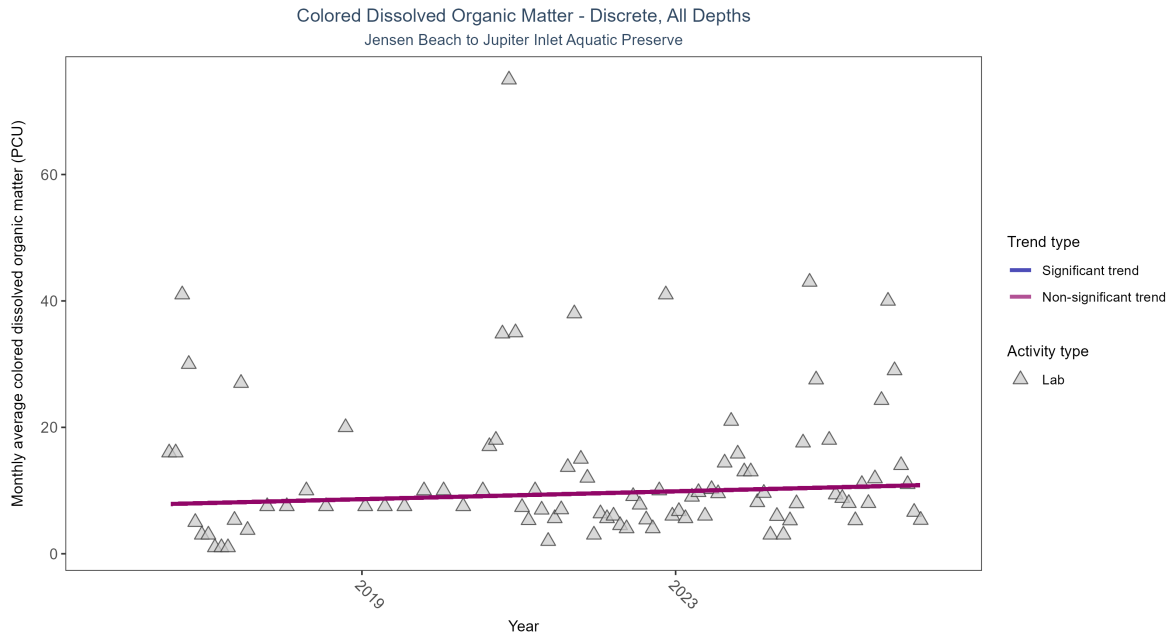


Figure 29: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 14: Colored dissolved organic matter showed no detectable trend between 2016 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	434	11	2016 - 2026	7.5	0.1522	7.70455	0.31061	0.0934

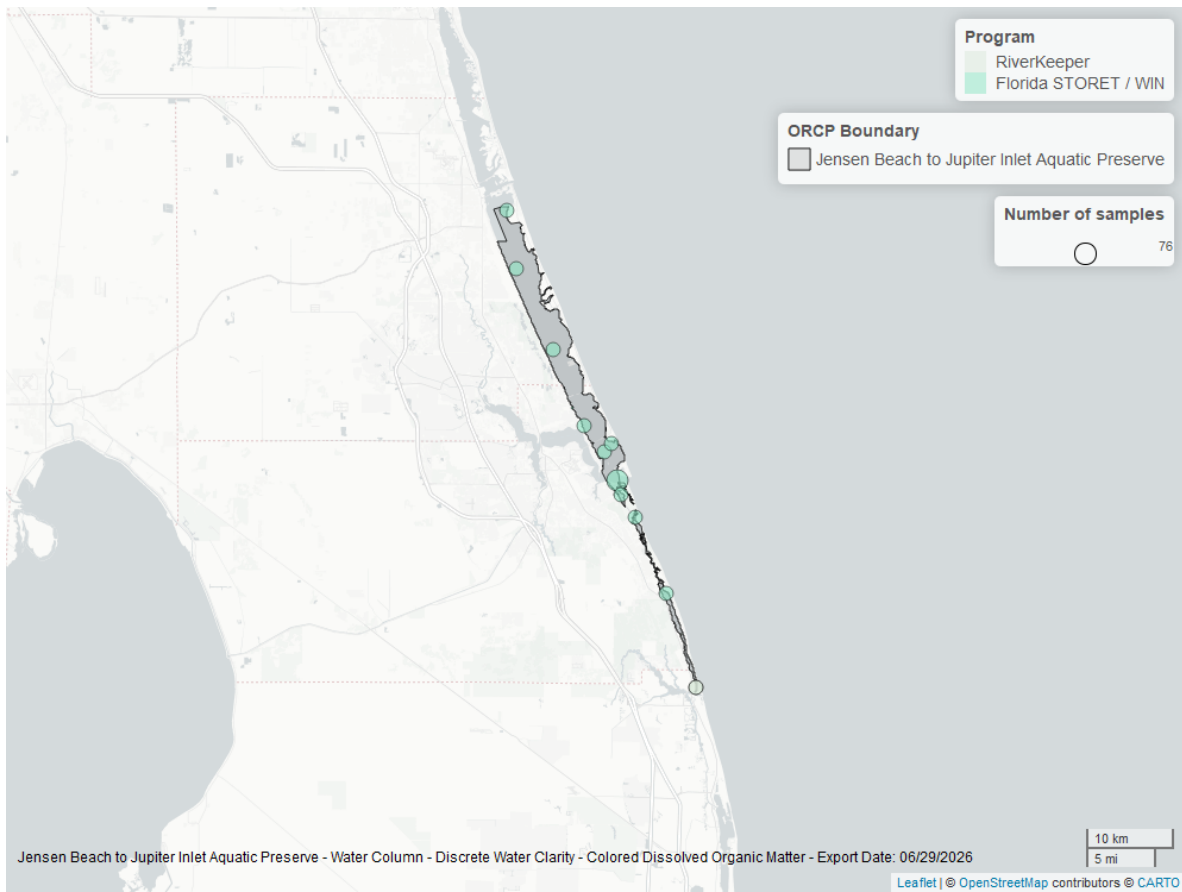


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.